Exact Stratified Presymplectic Reduction of the $SU(3)$-Gram Worldline Model

Independent calculation for the S^6 validation workbench

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Abstract

We compute, retaining $\alpha > 0$, $c > 0$, m_0, and every singular orbit type, the presymplectic reduction of

$$\int_I \Big(\langle \alpha Z^* Z, \dot{x} \rangle_L - \frac{e}{2} \big(\det(\alpha Z^* Z) - m_0^2 c^2 \big) \Big) \, \mathrm{d}\tau$$

by arbitrary τ-dependent left $SU(3)$ transformations. For $m_0 > 0$, the colour action is free and the reduced system is exactly the future-timelike first-order spinless relativistic particle; its physical phase space is $T^* \mathcal{H}^+_{m_0,c}$. For $m_0 = 0$, the nonzero rank-one physical quotient is $T^* \mathcal{N}^\times_+$, so its unparameterized null trajectories agree with the standard massless particle. The original multiplier system is nevertheless not exactly standard: the pulled-back constraint has zero differential on the massless locus, e drops out of the rank-one velocity equation, four real transverse directions are first-order null but not colour gauge, and at $Z = 0$ every $x(\tau)$ is stationary.

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1 Objects, constants, and equations

Let

$$\mathcal{J}_2 = \{A \in \mathrm{Mat}_{2\times 2}(\mathbb{C}) : A = A^*\}, \qquad \langle A, B\rangle_L = \frac{1}{2}\big(\mathrm{tr}\, A\, \mathrm{tr}\, B - \mathrm{tr}(AB)\big).$$

Then $\langle A, A\rangle_L = \det A$, and the pairing is nondegenerate. Write

$$\mathcal{P}_2 = \{p \in \mathcal{J}_2 : p \succeq 0\}, \quad \mathcal{P}_2^{++} = \{p \in \mathcal{J}_2 : p \succ 0\}, \quad \mathcal{N}_+^{\times} = \{p \in \mathcal{P}_2 : \mathrm{rank}_{\mathbb{C}}\, p = 1\}.$$

Retain

$$\mathscr{V} = \mathrm{Mat}_{3\times 2}(\mathbb{C}), \quad \alpha > 0, \quad c > 0, \quad m_0 \geq 0,$$

and define

$$P_\alpha(Z) = \alpha Z^* Z, \tag{1}$$

$$DP_{\alpha,Z}(\delta Z) = \alpha(\delta Z^* Z + Z^* \delta Z), \tag{2}$$

$$\Phi_{m_0,\alpha,c}(Z) = \frac{1}{2}\left(\det P_\alpha(Z) - m_0^2 c^2\right). \tag{3}$$

The history configuration space on a retained parameter interval I is

$$\mathfrak{C}_I = C^\infty(I, \mathcal{J}_2) \times C^\infty(I, \mathscr{V}) \times C^\infty(I, \mathbb{R}),$$

with coordinates (x, Z, e). Because the Lagrangian is first order in x, contains no $\dot{Z}$, and contains no $\dot{e}$, the corresponding instantaneous presymplectic phase carrier is

$$\mathcal{E} = \mathcal{J}_{2x} \times \mathscr{V}_Z \times \mathbb{R}_e.$$

Define

$$\theta = \langle P_\alpha(Z), \mathrm{d}x\rangle_L, \qquad \Omega = -\mathrm{d}\theta, \qquad \mathcal{H} = e\Phi_{m_0,\alpha,c}. \tag{4}$$

For tangent vectors $v = (v_x, v_Z, v_e)$ and $w = (w_x, w_Z, w_e)$,

$$\Omega(v, w) = \langle v_x, DP_{\alpha,Z}(w_Z)\rangle_L - \langle w_x, DP_{\alpha,Z}(v_Z)\rangle_L. \tag{5}$$

Proposition 1.1 (Exact presymplectic Euler–Lagrange equation)**.** *A path (x, Z, e) is stationary under compactly supported variations if and only if*

$$\det P_\alpha(Z) = m_0^2 c^2, \tag{6}$$

$$\frac{\mathrm{d}}{\mathrm{d}\tau} P_\alpha(Z) = 0, \tag{7}$$

$$\langle DP_{\alpha,Z}(\delta Z), \dot{x} - eP_\alpha(Z)\rangle_L = 0 \quad \textit{for every } \delta Z \in \mathscr{V}. \tag{8}$$

Equivalently, $\iota_{\dot\gamma}\Omega = \mathrm{d}\mathcal{H}$.

Proof. Since

$$D(\det)_p(A) = 2\langle p, A\rangle_L,$$

one has

$$\mathrm{d}\mathcal{H}(w) = \Phi_{m_0,\alpha,c} w_e + e\langle P_\alpha(Z), DP_{\alpha,Z}(w_Z)\rangle_L.$$

Insert (5) into $\iota_{\dot\gamma}\Omega = \mathrm{d}\mathcal{H}$ and test successively w_x, w_e, w_Z. This gives (7), (6), and (8). □

2 Colour quotient and orbit types

The left action is $U \cdot (x, Z, e) = (x, UZ, e)$.

Lemma 2.1 (Local colour is a zero-moment null symmetry)**.** *For $\xi \in \mathfrak{su}(3)$, let $\xi^{\#} = (0, \xi Z, 0)$. Then*

$$DP_{\alpha,Z}(\xi Z) = 0, \qquad \iota_{\xi^{\#}}\theta = 0, \qquad \iota_{\xi^{\#}}\Omega = 0.$$

Thus the zero map to $\mathfrak{su}(3)^*$ is a presymplectic moment map. The action is invariant under every τ-dependent $U : I \to SU(3)$, with no connection term.

Proof. Because $\xi^* = -\xi$,

$$DP_{\alpha,Z}(\xi Z) = \alpha(Z^*\xi^* Z + Z^*\xi Z) = 0.$$

The form identities follow from (4) and (5). Pointwise, $(UZ)^*(UZ) = Z^*Z$, even when U depends on τ. □

Theorem 2.2 (Gram quotient, fibres, and stabilizers)**.** *The map P_α induces a homeomorphism*

$$\mathscr{V}/SU(3) \xrightarrow{\sim} \mathcal{P}_2, \qquad [Z] \longmapsto P_\alpha(Z).$$

Its orbit types are

$\operatorname{rank}_{\mathbb{C}} Z$	*fibre*	$\operatorname{Stab}_{SU(3)}(Z)$
2	$SU(3)$	$\{I_3\}$
1	$SU(3)/SU(2) \cong S^5$	$SU(2)$
0	*a point*	$SU(3)$

The rank-two quotient is smooth and free. Rank one is a smooth nonprincipal orbit-type stratum with nontrivial stabilizer; the zero vertex is the singular apex of the total stratified quotient.

Proof. If $Z_1^* Z_1 = Z_2^* Z_2$, the map $Z_1 v \mapsto Z_2 v$ is an isometry between their images and extends to a unitary map of $\mathbb{C}^3$. Its action on the orthogonal complement can be adjusted to make its determinant one. In rank two the complement is one-dimensional and the adjustment is unique, hence the stabilizer is trivial. In rank one the stabilizer is $SU(2)$ on the two-dimensional complement. At zero it is all of $SU(3)$. Surjectivity has the retained section

$$Z_p = \alpha^{-1/2} \begin{pmatrix} p^{1/2} \\ 0 \quad 0 \end{pmatrix}.$$

Compactness of $SU(3)$ and continuity of the positive square root give the quotient homeomorphism. On $\mathcal{P}_2^{++}$ the section is smooth. On $\mathcal{N}_+^{\times}$, the rank-one identity

$$p^{1/2} = \frac{p}{\sqrt{\operatorname{tr} p}}$$

shows directly that the same section is smooth; together with the constant orbit type $SU(3)/SU(2)$, this gives the stated smooth nonprincipal stratum. □

Lemma 2.3 (Exact ranks of the Gram differential)**.** (i) *If $\operatorname{rank}_{\mathbb{C}} Z = 2$, then*

$$\operatorname{im} DP_{\alpha,Z} = \mathcal{J}_2, \qquad \ker DP_{\alpha,Z} = T_Z(SU(3) \cdot Z),$$

of real dimensions 4 and 8.

(ii) *If* $\operatorname{rank}_{\mathbb{C}} Z = 1$ *and* $p = P_\alpha(Z)$*, then*

$$\operatorname{im} DP_{\alpha,Z} = p^{\perp_L} = T_p \mathcal{N}_+^{\times}$$

has real dimension 3*. The full ambient kernel has dimension* 9*, but*

$$\ker(DP_{\alpha,Z}|_{T_Z \mathscr{V}_1}) = T_Z(SU(3) \cdot Z)$$

has dimension 5*, where* $\mathscr{V}_1 = \{Z : \operatorname{rank}_{\mathbb{C}} Z = 1\}$*. The remaining four real kernel directions are*

$$\operatorname{Hom}_{\mathbb{C}}\big(\ker Z, (\operatorname{im} Z)^{\perp}\big);$$

they raise rank and are not colour gauge.

(iii) *At* $Z = 0$*,* $DP_{\alpha,0} = 0$*.*

Proof. For rank two and arbitrary $A = A^*$, retain

$$\delta Z_A = \frac{1}{2\alpha} Z(Z^*Z)^{-1} A.$$

Substitution into (2) gives $DP_{\alpha,Z}(\delta Z_A) = A$. Hence the kernel has dimension $12 - 4 = 8$, equal to the free colour-orbit tangent.

For rank one write, without fixing either norm,

$$Z = ab^*, \qquad a \in \mathbb{C}^3 \setminus \{0\}, \quad b \in \mathbb{C}^2 \setminus \{0\}, \quad r = a^*a > 0.$$

Then $p = \alpha r b b^*$. If $w = \delta Z^* a$, every $w \in \mathbb{C}^2$ occurs by taking $\delta Z = a w^*/r$, and

$$\operatorname{im} DP_{\alpha,Z} = \{\alpha(wb^* + bw^*) : w \in \mathbb{C}^2\}.$$

The kernel of $w \mapsto wb^* + bw^*$ is $\{itb : t \in \mathbb{R}\}$, so the image has dimension three. It is contained in $p^{\perp_L}$, and equality follows by dimension.

Use the exact orthogonal splittings

$$\mathbb{C}^2 = (\ker Z)^{\perp} \oplus \ker Z, \qquad \mathbb{C}^3 = \operatorname{im} Z \oplus (\operatorname{im} Z)^{\perp}.$$

Retaining the nonzero singular value $\sigma > 0$, adapted unit vectors give

$$Z = \begin{pmatrix} \sigma & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \delta Z = \begin{pmatrix} a_0 & b_0 \\ c_1 & d_1 \\ c_2 & d_2 \end{pmatrix},$$

and

$$\alpha^{-1} DP_{\alpha,Z}(\delta Z) = \begin{pmatrix} 2\sigma \operatorname{Re} a_0 & \sigma b_0 \\ \sigma \overline{b_0} & 0 \end{pmatrix}.$$

The ambient kernel has $\operatorname{Re} a_0 = b_0 = 0$ and dimension nine. The linearized rank-one equations also impose $d_1 = d_2 = 0$. The five remaining directions are the $SU(3)/SU(2)$-orbit tangent; the d_1, d_2 block is the displayed complex Hom-space. The zero case follows from (2). □

3 Forms after colour reduction

Define

$$\overline{\mathcal{E}} = \mathcal{J}_{2x} \times \mathcal{P}_{2p} \times \mathbb{R}_e \cong \mathcal{E}/SU(3), \qquad q(x, Z, e) = (x, P_\alpha(Z), e).$$

Proposition 3.1 (Reduced one-form and two-form)**.** *The forms are basic for the colour action and satisfy*

$$\theta = q^*\overline{\theta}, \qquad \Omega = q^*\overline{\Omega},$$

where

$$\overline{\theta} = \langle p, \mathrm{d}x\rangle_L, \tag{9}$$

$$\overline{\Omega} = -\mathrm{d}\overline{\theta}, \tag{10}$$

$$\overline{\Omega}((v_x, v_p), (w_x, w_p)) = \langle v_x, w_p\rangle_L - \langle w_x, v_p\rangle_L. \tag{11}$$

After omitting the multiplier coordinate, this is symplectic on $\mathcal{J}_2 \times \mathcal{P}_2^{++}$. On $\mathcal{J}_2 \times \mathcal{N}_+^\times$, its kernel is exactly

$$\ker \overline{\Omega}_{(x,p)} = \mathrm{span}_{\mathbb{R}}\{(p, 0)\}.$$

On $\mathcal{J}_2 \times \{0\}$, the stratum-restricted two-form is zero.

Proof. Basicness follows from Lemma 2.1; the formulas follow by differentiation. The rank-two assertion uses that $\mathcal{P}_2^{++}$ is open in $\mathcal{J}_2$. At nonzero rank-one p, one has

$$T_p\mathcal{N}_+^\times = p^{\perp_L}.$$

If (v_x, v_p) annihilates (11), testing arbitrary w_x gives $v_p = 0$, and testing $w_p \in p^{\perp_L}$ gives $v_x \in (p^{\perp_L})^{\perp_L} = \mathrm{span}_{\mathbb{R}}\{p\}$. At $p = 0$, the zero stratum has no momentum tangent direction. □

4 Positive mass

For $m_0 > 0$, define the retained future shell

$$\mathcal{H}^+_{m_0,c} = \{p \in \mathcal{P}_2^{++} : \det p = m_0^2c^2\}.$$

The parameter α remains present in the exact quotient map $p = \alpha Z^*Z$; the displayed subset of momentum space itself depends on m_0, c.

Theorem 4.1 (Exact massive reduction)**.** *Let $m_0 > 0$. Then:*

(a) *The constraint surface*

$$\Sigma_{m_0,\alpha,c} = \mathcal{J}_2 \times \{Z : \det P_\alpha(Z) = m_0^2c^2\} \times \mathbb{R}_e$$

is smooth and the colour action on it is free.

(b) *Its colour quotient is*

$$\overline{\Sigma}_{m_0,c} = \mathcal{J}_2 \times \mathcal{H}^+_{m_0,c} \times \mathbb{R}_e.$$

Before colour reduction the kernel of the restricted two-form is the direct sum of the eight-dimensional colour-orbit tangent, $\mathrm{span}_{\mathbb{R}}\{(p, 0)\}$, and $\mathrm{span}_{\mathbb{R}}\{\partial_e\}$. After colour reduction and omission of e, only $\mathrm{span}_{\mathbb{R}}\{(p, 0)\}$ remains.

(c) *The equations are*

$$\dot{p} = 0, \qquad \dot{x} = e\,p, \qquad \det p = m_0^2 c^2,$$

while $\dot{e}$ and motion in the colour fibre are arbitrary.

(d) *The physical phase space*

$$\mathcal{R}_{m_0,c} = \frac{\mathcal{J}_2 \times \mathcal{H}^+_{m_0,c}}{(x,p) \sim (x + sp, p),\ s \in \mathbb{R}}$$

is canonically symplectomorphic to $T^*\mathcal{H}^+_{m_0,c}$.

(e) *The colour-reduced positive-mass model is exactly the standard future-directed spinless first-order relativistic particle of mass m_0 and speed constant c.*

Proof. The constraint implies $\operatorname{rank}_{\mathbb{C}} Z = 2$. By Lemma 2.3, P_α is a submersion there, and

$$D(\det)_p = 2\langle p, \cdot \rangle_L \neq 0.$$

This proves smoothness. Theorem 2.2 gives freeness and the colour quotient. Since

$$T_p\mathcal{H}^+_{m_0,c} = p^{\perp_L},$$

the proof of Proposition 3.1 gives the characteristic line. Pullback through the Gram submersion adds exactly the colour-orbit kernel, and the multiplier adds ∂_e.

Surjectivity of $DP_{\alpha,Z}$ turns (8) into $\dot{x} = ep$; the other equations are (6) and (7). A path with constant p lies in one free Gram fibre and differs locally from a constant lift by a τ-dependent colour transformation.

For the physical identification, if $A \in T_p\mathcal{H}^+_{m_0,c}$, define

$$\xi_{[x],p}(A) = -\langle x, A \rangle_L.$$

This is unchanged by $x \mapsto x + sp$. Conversely, the annihilator of $p^{\perp_L}$ is $\operatorname{span}_{\mathbb{R}}\{p\}$, so every cotangent covector has one and only one class $[x]$. Hence

$$([x], p) \longmapsto (p, \xi_{[x],p})$$

is a fibrewise linear bijection to $T^*\mathcal{H}^+_{m_0,c}$. The characteristic $\mathbb{R}$-action is free because every shell momentum is nonzero. It is proper: choose any Euclidean norm on $\mathcal{J}_2$. For compact sets $K_1, K_2 \subset \mathcal{J}_2 \times \mathcal{H}^+_{m_0,c}$, the momenta occurring in them have a positive minimum Euclidean norm δ; if $(x,p) \in K_1$ and $(x+sp,p) \in K_2$, boundedness of the two position projections gives $|s|\delta \leq C(K_1, K_2)$. Thus the orbit space is smooth. The smooth vector-bundle map

$$(x,p) \longmapsto \big(p,\ A \mapsto -\langle x, A \rangle_L\big)$$

is fibrewise onto with kernel $\operatorname{span}_{\mathbb{R}}\{p\}$, and therefore descends with a smooth inverse. The displayed bijection is consequently a diffeomorphism.

The basic one-form on the shell is

$$\theta_{\text{basic}} = -\langle x, \mathrm{d}p \rangle_L. \tag{12}$$

It is the pullback of the tautological cotangent one-form, and

$$\overline{\theta} = \mathrm{d}\langle p, x \rangle_L + \theta_{\text{basic}}. \tag{13}$$

Therefore $\overline{\Omega} = -\mathrm{d}\theta_{\mathrm{basic}}$, the canonical cotangent symplectic form under the convention $\omega_{\mathrm{can}} = -\mathrm{d}\lambda_{\mathrm{can}}$. The original $\overline{\theta}$ is not horizontal on the massive characteristic because $\overline{\theta}(p,0) = m_0^2c^2$; the exact boundary term in (13) produces the reduced one-form.

Finally the reduced action is

$$\int_I \Big(\langle p, \dot{x} \rangle_L - \frac{e}{2}(\det p - m_0^2 c^2) \Big)\, \mathrm{d}\tau.$$

Because $\mathcal{P}_2^{++}$ is open in $\mathcal{J}_2$, p has every infinitesimal Hermitian variation. This is the standard covariant first-order spinless particle action on the future-timelike sector, with no residual internal symplectic factor. □

Proposition 4.2 (Exact Hamiltonian gauge quasi-symmetry)**.** *For a smooth real $\varepsilon(\tau)$, set*

$$\delta_\varepsilon x = \varepsilon P_\alpha(Z), \qquad \delta_\varepsilon Z = 0, \qquad \delta_\varepsilon e = \dot{\varepsilon}.$$

Then

$$\delta_\varepsilon L = \frac{\mathrm{d}}{\mathrm{d}\tau}\left[\frac{\varepsilon}{2}\big(\det P_\alpha(Z) + m_0^2 c^2\big) \right]. \tag{14}$$

For compactly supported ε this is an exact gauge quasi-symmetry, whose phase-space part generates the shell characteristic where that characteristic is regular.

Proof. Put $p = P_\alpha(Z)$ and $q = \det p = \langle p, p \rangle_L$. Since $\dot{q} = 2\langle p, \dot{p} \rangle_L$,

$$\begin{aligned} \delta_\varepsilon L &= \langle p, \dot{\varepsilon} p + \varepsilon \dot{p} \rangle_L - \frac{\dot{\varepsilon}}{2}(q - m_0^2 c^2) \\ &= \frac{\dot{\varepsilon}}{2}(q + m_0^2 c^2) + \frac{\varepsilon}{2}\dot{q}, \end{aligned}$$

which is (14). □

5 Massless rank one

Set

$$\mathcal{C}_{0,\alpha,c} = \{ Z : \det P_\alpha(Z) = 0 \} = \{ Z : \operatorname{rank}_{\mathbb{C}} Z \le 1 \}.$$

The constant $c > 0$ is retained in the model, although it cancels from the set-theoretic $m_0 = 0$ equation.

Lemma 5.1 (The pulled-back massless constraint is irregular)**.** *Writing the rows of Z as $Z_{a\bullet}$,*

$$\det P_\alpha(Z) = \alpha^2 \sum_{1 \le a < b \le 3} |Z_{a1} Z_{b2} - Z_{a2} Z_{b1}|^2. \tag{15}$$

Consequently,

$$\mathrm{d}(\det P_\alpha)_Z = 0 \quad \textit{for every } Z \in \mathcal{C}_{0,\alpha,c}.$$

The single real multiplier equation describes the rank-at-most-one set, but its differential supplies none of the four real normal equations on the smooth rank-one stratum.

Proof. Equation (15) is Cauchy–Binet with the factor α^2 retained. On its zero set every complex minor vanishes, so differentiation of the sum of squared absolute values gives zero. Equivalently, at rank one,

$$\mathrm{d}(\det P_\alpha)_Z(\delta Z) = 2\langle p, DP_{\alpha,Z}(\delta Z) \rangle_L = 0$$

by Lemma 2.3; at $Z = 0$ use $DP_{\alpha,0} = 0$. □

Theorem 5.2 (Nonzero massless equations and physical quotient)**.** *Suppose* $m_0 = 0$ *and* $\operatorname{rank}_{\mathbb{C}} Z = 1$*, and put* $p = P_\alpha(Z) \in \mathcal{N}_+^\times$*. Then:*

(a) *The colour fibre is* $SU(3)/SU(2)$*, with stabilizer* $SU(2)$*.*

(b) *The complete equations are*

$$\dot{p} = 0, \qquad \dot{x} = \lambda(\tau)p \quad \text{for some real } \lambda(\tau), \qquad e(\tau) \text{ arbitrary.} \tag{16}$$

No equation identifies λ *with the written multiplier* e*.*

(c) *On the intrinsic smooth rank-one shell, the kernel before colour reduction is the direct sum of*

$$T_Z(SU(3) \cdot Z), \qquad \operatorname{span}_{\mathbb{R}}\{(p, 0)\}, \qquad \operatorname{span}_{\mathbb{R}}\{\partial_e\}.$$

After colour reduction and omission of e*, only the characteristic* $\operatorname{span}_{\mathbb{R}}\{(p, 0)\}$ *remains.*

(d) *The physical phase space*

$$\mathcal{R}_0^\times = \frac{\mathcal{J}_2 \times \mathcal{N}_+^\times}{(x, p) \sim (x + sp, p)}$$

is canonically symplectomorphic to $T^*\mathcal{N}_+^\times$*, with reduced one-form* $-\langle x, \mathrm{d}p\rangle_L$ *and its canonical two-form. No quotient by positive rescaling of* p *occurs.*

(e) *Thus the unparameterized spacetime curves and physical symplectic quotient agree with the standard future-directed nonzero spinless massless particle. The original triples* (x, Z, e)*, their off-shell neighbourhood, and their multiplier equations are not exactly the standard massless system.*

Proof. The fibre statement is Theorem 2.2. By Lemma 2.3, equation (8) says

$$\dot{x} - ep \in (p^{\perp_L})^{\perp_L} = \operatorname{span}_{\mathbb{R}}\{p\}.$$

Since ep lies in this line, this is equivalent to $\dot{x} = \lambda p$, with no relation between λ and e. Equation (7) gives $\dot{p} = 0$. A path in a fixed rank-one Gram fibre differs locally from a constant representative by a τ-dependent $SU(3)$ transformation; the $SU(2)$ stabilizer makes this gauge description reducible.

On the intrinsic rank-one manifold, Lemma 2.3 identifies the vertical kernel with the five colour directions. Proposition 3.1 adds the one spacetime characteristic, and e adds ∂_e.

Since $T_p\mathcal{N}_+^\times = p^{\perp_L}$, the massive cotangent construction applies without division by $\det p$:

$$([x], p) \longmapsto (p,\ A \longmapsto -\langle x, A\rangle_L)\,.$$

Every $p \in \mathcal{N}_+^\times$ is nonzero. The freeness, properness, and smooth vector-bundle argument in the massive proof applies to compact subsets of $\mathcal{J}_2 \times \mathcal{N}_+^\times$ verbatim, and proves that this map is a diffeomorphism to $T^*\mathcal{N}_+^\times$. Moreover, (13) identifies the canonical form. Reparametrizing the worldline changes x along p; it does not identify distinct positive magnitudes of p. Lemma 5.1 and the absence of $\lambda = e$ establish the final inequivalence statement. □

Counterexample 5.3 (Rank-one multiplier mismatch)**.** *Retain arbitrary* $s > 0$*, and set*

$$Z_s = \begin{pmatrix} s & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad p_s = \alpha \begin{pmatrix} s^2 & 0 \\ 0 & 0 \end{pmatrix}.$$

The path

$$Z(\tau) = Z_s, \qquad x(\tau) = \tau p_s, \qquad e(\tau) = 0$$

satisfies all equations in Proposition 1.1 for $m_0 = 0$. The standard action with momentum independently varied in $\mathcal{J}_2$ and the same $e = 0$ would require $\dot{x} = ep_s = 0$.

Under Proposition 4.2, a solution $\dot{x} = \lambda p$, $\dot{p} = 0$, transforms by

$$\lambda \longmapsto \lambda + \dot{\varepsilon}, \qquad e \longmapsto e + \dot{\varepsilon}.$$

Thus $\lambda - e$ is invariant under the stated Hamiltonian gauge; the discrepancy remains when the original multiplier is retained.

Counterexample 5.4 (Four non-colour first-order null directions)**.** *Retain arbitrary $s > 0$ and $t \in \mathbb{C} \setminus \{0\}$, and put*

$$Z_s = \begin{pmatrix} s & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \Delta_t = \begin{pmatrix} 0 & 0 \\ 0 & t \\ 0 & 0 \end{pmatrix}.$$

Then

$$DP_{\alpha,Z_s}(\Delta_t) = 0, \qquad \det P_\alpha(Z_s + \varepsilon\Delta_t) = \alpha^2 s^2 |t|^2 \varepsilon^2 \quad (\varepsilon \in \mathbb{R}). \tag{17}$$

The vector Δ_t is not tangent to a colour orbit, since every ξZ_s has zero second column. It is not tangent to the rank-one variety, because the perturbed matrix has rank two for $\varepsilon \neq 0$. Allowing both complex coordinates in $(\operatorname{im} Z_s)^\perp$ gives the four real transverse directions of Lemma 2.3. They are second-order shell-opening directions, not gauge.

6 The zero-momentum vertex

Theorem 6.1 (Additional singular branch)**.** *Let $m_0 = 0$ and $Z = 0$. Then $p = 0$, the stabilizer is all of $SU(3)$, and both θ and Ω vanish on $\mathcal{J}_2 \times \{0\} \times \mathbb{R}_e$. Every pair of paths*

$$x : I \to \mathcal{J}_2, \qquad e : I \to \mathbb{R},$$

with $Z(\tau) \equiv 0$, satisfies the Euler–Lagrange equations.

The phase part of the Hamiltonian gauge vanishes at $p = 0$, while the presymplectic kernel contains every x-direction. Quotienting by the actual colour and Hamiltonian-gauge orbits therefore leaves the whole $\mathcal{J}_2$ branch, whereas quotienting by all characteristic leaves would collapse it to one point. This is an exact failure of regular Dirac reduction at the vertex.

Proof. At $Z = 0$, one has $p = 0$ and $DP_{\alpha,0} = 0$. Every equation in Proposition 1.1 is identically satisfied, independent of $\dot{x}, e, \dot{e}$. The form and gauge statements follow from (4) and $\delta_\varepsilon x = \varepsilon p = 0$. □

Counterexample 6.2 (Stationary zero momentum with nonconstant position)**.** *The retained paths*

$$x(\tau) = \begin{pmatrix} \tau^2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Z(\tau) = 0, \qquad e(\tau) = \sin\tau$$

are stationary for the present $m_0 = 0$ action. The standard independent-momentum action gives $\dot{x} = ep = 0$ at $p = 0$, and therefore excludes this path. Removing $p = 0$, as is conventional for physical massless phase space, removes this vertex counterexample but does not regularize the off-shell rank-one constraint.

7 Kernel dimensions and exact verdict

Proposition 7.1 (Every kernel jump)**.** *Omit the always-null multiplier direction for the table. On the ambient* 16*-real-dimensional* $\mathcal{J}_{2x} \times \mathcal{V}_Z$*,*

$\operatorname{rank}_{\mathbb{C}} Z$	$\operatorname{rank}_{\mathbb{R}} DP_{\alpha,Z}$	$\operatorname{rank}_{\mathbb{R}} \Omega$	$\dim_{\mathbb{R}} \ker \Omega$
2	4	8	8
1	3	6	10
0	0	0	16

On the 12*-real-dimensional intrinsic rank-one stratum* $\mathcal{J}_{2x} \times \mathcal{V}_1$*, the pulled-back form has rank* 6 *and nullity* 6*: five colour directions and one spacetime characteristic.*

Proof. Using the Lorentz pairing to identify $\mathcal{J}_2 \cong \mathcal{J}_2^*$, the matrix of Ω is

$$\begin{pmatrix} 0 & G_L DP_{\alpha,Z} \\ -(DP_{\alpha,Z})^{\mathsf{T}} G_L & 0 \end{pmatrix}.$$

Its rank is twice the rank of $DP_{\alpha,Z}$. Apply Lemma 2.3 in the ambient space and on $T_Z \mathcal{V}_1$. □

Theorem 7.2 (Final comparison)**.** *The strongest correct result is:*

(1) *for* $m_0 > 0$*, colour reduction gives exactly the standard future-timelike first-order spinless particle, including its equations, constraint characteristic, and physical cotangent symplectic space;*

(2) *for* $m_0 = 0$ *and* $p \neq 0$*, colour and characteristic reduction gives exactly the standard future-null physical phase space and unparameterized null trajectories;*

(3) *the full* $m_0 = 0$ *pulled-back multiplier action is not the standard massless variational system: its constraint has zero differential, its rank-one equation is* $\dot{x} = \lambda p$ *rather than* $\dot{x} = ep$*, its ambient kernel contains four non-colour transverse directions, and its vertex admits arbitrary* $x(\tau)$*.*

The exact relating morphism is $P_\alpha(Z) = \alpha Z^* Z$*; the exact failure locus is the singular differential of* $\det \circ P_\alpha$ *on the rank-at-most-one variety.*

Proof. Combine Theorem 4.1, Theorem 5.2, Lemma 5.1, and Counterexamples 5.3–6.2. □

8 Deterministic verification companion

The companion program `verify_worldline_reduction.py` checks the determinant polarization, the rank-two Gram right inverse, all Gram and presymplectic ranks, all colour-orbit ranks, the intrinsic rank-one tangent calculation, the zero gradient and quadratic transverse opening of the massless constraint, the cotangent-form sign, and (14), using exact symbolic arithmetic.

A Comparison with the extended exact Lorentz packet

This appendix records a bounded line-specific audit of source lines 875–1654 of `exact_lorentz_observation_bridge.tex`, file SHA-256

`2BE16E1DC71B2E5DC04238F0FBE440138258D0825A4841AB2F29662587681599`.

That packet adds a third column $\phi \in \mathbb{C}^3$, writes $B = [Z\ \phi]$, and uses

$$G = Z^* Z, \quad z = Z^* \phi, \quad \lambda = \phi^* \phi, \quad \omega = \det B.$$

Its mass convention is $m^2_{\mathrm{packet}} = \det p$, with physical mass $M_{\mathrm{phys}} = \sqrt{\det p}/c$. The convention in the audited S^6 module is $m^2_{\alpha,c} = \det p / c^2$. The exact dictionary is therefore

$$m^2_{\mathrm{packet}} = c^2 m^2_{\alpha,c}, \qquad M_{\mathrm{phys}} = m_{\alpha,c};$$

there is no mathematical disagreement.

A.1 Claim ledger

Source lines	Verdict	Exact scope
875–950	validated	$p = \alpha Z^* Z$ is a proper surjection, its fibres are the $SU(3)$-orbits, and $\det p = \alpha^2 \sum_{a<b} \lvert\Delta_{ab}(Z)\rvert^2$ has range $[0, \infty)$. The zero fibre is rank at most one.
952–1027	validated as a multiplication-operator model	With the stated pushforward measure, the observer-energy and mass-squared multiplication operators have essential ranges $[0, \infty)$, no point spectrum, and no positive gap after a vacuum is adjoined. This is a kinematic/free spectral statement, not a statement about an interacting field Hamiltonian.
1029–1476	validated	For $\lambda_0 > 0$, $\omega_0 \neq 0$, the constrained quotient, associated-ball description, momentum fibres, and sharp lower bound $\det p \geq \alpha^2 \lvert\omega_0\rvert^2/\lambda_0$ are algebraic and differential-topological statements. Since $\omega_0 \neq 0$ forces B invertible and Z rank two, the massless rank-one defect proved above does not invalidate these claims.
1482–1531	valid invariance and constraints, incomplete dynamics	The displayed action is invariant and its multiplier and x variations give the stated constraints and $\dot{p} = 0$. Its omitted (Z, ϕ) equations materially change the dynamical interpretation; Theorem A.1 below computes them intrinsically.
1533–1603	validated with scope qualification	The multiplication spectra and the direct-sum/Fock vacuum gap equal the displayed threshold. The threshold is inserted by fixing $\lambda = \lambda_0$ and $\omega = \omega_0 \neq 0$; it is not a dynamically generated interacting mass gap and is not a Yang–Mills mass-gap theorem.
1605–1633	validated and scope-sharpening	The two explicit degenerating sequences prove that controlling only one of λ and ω gives no positive threshold. Thus the positive lower endpoint is exactly constraint-selected.
1635–1653	validated	The diagonal boost excludes a positive-definite quadratic form on the spinor carrier invariant under all of $SL_2(\mathbb{C})$.

A.2 The full dynamics of the packet's constrained action

Retain the packet's constants

$$\lambda_0 > 0, \qquad \omega_0 \in \mathbb{C}^*, \qquad \alpha > 0$$

and its smooth colour quotient

$$\mathcal{Q}_{\lambda_0,\omega_0} \cong \mathrm{Herm}_2^0(\mathbb{C}) \times \{w \in \mathbb{C}^2 : \|w\|^2 < \lambda_0\}.$$

Put

$$\mu(w) = \frac{|\omega_0|}{\sqrt{\lambda_0 - \|w\|^2}}, \qquad G(A,w) = \mu(w)e^A, \qquad p(A,w) = \alpha G(A,w). \tag{18}$$

Eliminating the three regular multiplier constraints from source lines 1484–1495 gives the intrinsic action

$$S_{\text{packet}}[x,A,w] = \int_I \langle p(A,w), \dot{x}\rangle_L \, \mathrm{d}\tau \tag{19}$$

on $\mathcal{J}_{2x} \times \mathcal{Q}_{\lambda_0,\omega_0}$. The use of the intrinsic constrained action loses no Euler–Lagrange equation: the constraint map is a submersion on $\omega_0 \neq 0$, so the Lagrange-multiplier theorem identifies ambient stationarity with stationarity against every tangent vector of the constrained manifold. This is a statement about classical stationary paths; it does not assert equality of quantum measures or path integrals after formally integrating out the multipliers.

Theorem A.1 (Complete reduced dynamics of the constrained packet)**.** *For the momentum projection in* (18)*:*

(i) *If $w \neq 0$, then $Dp_{(A,w)}$ has real rank 4, and its kernel has dimension 3, tangent to the S^3 momentum fibre. The Euler–Lagrange equations of* (19) *are*

$$\dot{p} = 0, \qquad \dot{x} = 0,$$

with motion along the fixed S^3 fibre presymplectically null.

(ii) *If $w = 0$, then*

$$\det p = \frac{\alpha^2|\omega_0|^2}{\lambda_0},$$

$Dp_{(A,0)}$ has rank 3, its image is $p^{\perp_L}$, and its kernel consists of all four real w-directions. The pointwise equations are

$$\dot{p} = 0, \qquad \dot{x} = \lambda(\tau)p.$$

Every actual solution remaining on the threshold has constant (A,w), because the momentum fibre over a threshold momentum is a single point.

(iii) *Consequently, the source action is not a family of ordinary massive particle actions over all masses above the lower endpoint. Only its threshold stratum has the ordinary worldline characteristic; strict-interior configurations have $\dot{x} = 0$. The observer multiplication operator of source lines 1533–1603 is a separate positive-energy representation and is not the Hamiltonian generating τ-evolution in* (19)*.*

Proof. For fixed w, varying A varies p through the three-dimensional fixed-determinant future hyperboloid, so the image of the A-differential is $p^{\perp_L}$. Direct differentiation without suppressing any constant gives

$$D\mu_w(\delta w) = \frac{|\omega_0|\,\mathrm{Re}\langle w, \delta w\rangle}{(\lambda_0 - \|w\|^2)^{3/2}}. \tag{20}$$

When $w \neq 0$, a radial δw makes (20) nonzero and changes $\det p = \alpha^2\mu(w)^2$. This supplies a fourth direction transverse to $p^{\perp_L}$. The remaining w-directions satisfy $\mathrm{Re}\langle w, \delta w\rangle = 0$, form a three-dimensional space, and are exactly tangent to the S^3 momentum fibre.

At $w = 0$, equation (20) vanishes in every direction, so the four w-directions lie in $\ker Dp$ and the image is only $p^{\perp_L}$. These directions open the mass above its minimum at second order:

$$\det p(A, \varepsilon v) = \frac{\alpha^2|\omega_0|^2}{\lambda_0 - \varepsilon^2\|v\|^2}.$$

For an action $\int \langle p(q), \dot{x} \rangle_L \mathrm{d}\tau$, variation of x gives $\dot{p} = 0$, while variation of q gives

$$\langle Dp_q(\delta q), \dot{x} \rangle_L = 0 \quad \text{for all } \delta q.$$

If $w \neq 0$, the image is all of $\mathcal{J}_2$, hence $\dot{x} = 0$. If $w = 0$, the image is $p^{\perp_L}$, hence $\dot{x} \in \mathrm{span}_{\mathbb{R}}\{p\}$. The packet's own momentum-fibre theorem states that the fibre over a threshold momentum is a singleton, which completes the solution statement. □

A.3 Relation to the rank-one irregularity

The packet's fixed-$\omega_0 \neq 0$ constrained manifold never meets $\mathrm{rank}_{\mathbb{C}} Z \leq 1$, so the zero differential of $\det \circ P_\alpha$ does not invalidate its topology or positive lower bound. The exact relation is instead:

1. on the unrestricted carrier, the rank-one and zero strata give the massless irregularities proved in Theorems 5.2 and 6.1;

2. on the packet's constrained carrier, Z remains rank two, but the momentum projection develops a different four-real-dimensional critical kernel at its positive-mass threshold $w = 0$;

3. the algebraic inequality is a sharp, externally constraint-selected threshold; a dynamically generated gap would require a specified interacting Hamiltonian or field operator whose spectrum produces the scale rather than receiving λ_0, ω_0 as fixed input.